

BDA400 - Data Science Tools and Techniques

Assignment 5

Technical Analysis using R, Development Phase

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Repository location:

<https://github.com/nnediokafor/TechnicalAnalysis/tree/main/Assignment5>

Submission filename: Okafor_Nnedi_CA_BDA400_A05.pdf

1. Overview

This assignment implements the required technical analysis indicators in R using only base R functions. No external technical-analysis libraries are used. Each indicator is placed in a separate R script file and follows the function names and logic described in the assignment templates and pseudocode.

2. How to Run the Code

1. Open RStudio.
2. Download or clone the Assignment5 folder from the GitHub repository.
3. Open the required .R script in RStudio.
4. Click Source in the top-right of the script window, or press Ctrl+Shift+Enter to run the complete script.
5. Each script contains a small example at the bottom. The output will appear in the Console.
6. For MACD and Stochastic RSI, the helper calculations are included in the same script so the file can run on its own.

3. Implementation Explanations

Simple Moving Average (SMA)

File: sma.R

The sma() function calculates the average of each rolling window of values for a chosen period. It checks that the data is numeric, confirms that the period is valid, and then loops through the data. For each window, the values are added and divided by the period. The function returns the sequence of SMA values.

Exponential Moving Average (EMA)

File: ema.R

The ema() function gives more weight to recent observations. It calculates the multiplier as $2 / (\text{period} + 1)$. The first EMA value is the first data value. Each later value is calculated from the current data point, the previous EMA, and the multiplier. The function returns one EMA value for each input value.

Moving Average Convergence Divergence (MACD)

File: macd.R

The macd() function calculates a short-period EMA and a long-period EMA, then subtracts the long EMA from the short EMA to get the MACD line. It calculates another EMA of the MACD line to create the signal line. The histogram is the MACD line minus the signal line. The function returns all three results in a list.

Standard Deviation

File: stdev.R

The stdev() function follows the population standard deviation formula provided in the assignment. It calculates the mean, subtracts the mean from each observation, squares the differences, finds their average, and takes the square root of the variance. The function returns one standard deviation value.

Linear Regression

File: linreg.R

The `linreg()` function selects the required portion of the source data using the regression length and offset. It creates index values, calculates their means, computes the regression numerator and denominator, and then calculates the slope and intercept. Finally, it uses the regression equation to create predicted values and returns the slope, intercept, and predictions.

Relative Strength Index (RSI)

File: rsi.R

The `rsi()` function first calculates changes between consecutive values. Positive changes are stored as gains and negative changes are stored as losses. Average gains and losses are calculated for the selected period, then Wilder-style smoothing is applied. The function converts relative strength into RSI values between 0 and 100.

Stochastic RSI (StochRSI)

File: stoch_rsi.R

The `stoch_rsi()` function first calculates RSI values. It then normalizes the valid RSI values using the minimum and maximum RSI values. A moving average is used to calculate the %K line and another moving average is used to calculate the %D line. The function returns both lines in a list.

Crossover

File: crossover.R

The `crossover()` function compares two arrays of the same length. It checks whether the first array moves from below the second array to above it, or from above to below it. The result records Up, Down, or None for each observation, matching the assignment pseudocode.

Crossunder

File: crossunder.R

The `crossunder()` function checks when the first array moves from greater than or equal to the second array to less than it. It returns True for a crossunder event and False when no event occurs, with None used for the first observation because there is no previous value to compare.

4. Testing and Accuracy

Each script contains sample data at the bottom so that the function can be run immediately in RStudio. The tests are designed to confirm that each function returns an output of the expected type and that the calculations follow the supplied formulas and pseudocode. Basic input checks are also included where appropriate to prevent invalid periods, empty data, non-numeric input, or arrays of different lengths from producing misleading results.

5. Files Submitted

sma.R

ema.R

macd.R

stdev.R

linreg.R

rsi.R

stoch_rsi.R

crossover.R

crossunder.R

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